LESSON GUIDE

Technology Plus

Unit 4: Networking

6.5.1 – Securing Small Wireless Networks

Lesson Overview:

In this lesson, students will learn how to identify and adjust wireless router settings to strengthen network security and prevent common vulnerabilities.

Students will:

- Describe common wireless security settings such as SSID, PSK, encryption types, WPS, and firewall.
- Explain how each setting affects the security of a small wireless network.
- Use best practices to evaluate whether a router's settings are secure.

Guiding Question: How can I tell if a wireless router's settings are secure, and what can I do to fix it if they're not?

Suggested Grade Levels: 8-12

Technology Needed: Yes

Approximate Time Required: 60-75 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 6.5 – Given a scenario, configure security settings for a small wireless network.
  - Changing the services set identifier (SSID)
  - Changing the default password
  - Encrypted vs. unencrypted
    - Open
    - Pre-shared key
    - Wireless Protected Access (WPA)
    - Wireless Protected Access 2 (WPA2)
    - Wireless Protected Access 3 (WPA3)

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Lesson Background

Background Information:

Most people only touch their home router twice: once to plug it in and once to panic when the Wi-Fi goes out. That's what makes this lesson so critical — it walks students through the actual settings that protect a network from being the low-hanging fruit on a hacker's hit list.
Students won't become IT administrators overnight, but they will come away with the confidence to change a default password, recognize outdated encryption, and know why features like WPS are more risk than reward.

It's easy to click the wrong button and knock everyone offline — or worse, to not click anything and leave the door wide open. This lesson gives students a safe space to explore those settings before the pressure's on.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 6.5.1 – Securing Small Wireless Networks • Lab Slides 6.5.1 – Wi-Fortify: Secure the Router
Materials per Student	• Student Notes 6.5.1 – Securing Small Wireless Networks • Student Handout 6.5.1 – Securing Small Wireless Networks • Assessment 6.5.1 – Securing Small Wireless Networks (optional)

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Lesson Background

Cyber Terms:

Term	Definition
service set identifier (SSID)	the name of a Wi-Fi network
Wi-Fi Protected Access (WPA)	a security standard that protects Wi-Fi networks by encrypting the data sent between devices and the router
pre-shared key (PSK)	a shared Wi-Fi password used by all devices on a network
Wi-Fi Protected Setup (WPS)	a feature that lets you connect devices to a Wi-Fi network without typing a password – usually by pressing a button on the router or entering a short PIN
guest network	a separate Wi-Fi network for visitors that keeps them isolated from your main device and files
firmware	software built into a device that controls how its hardware works – usually stored on a chip and updated through the device's settings to fix bugs or improve security
forward secrecy	a security feature of WPA3 that uses temporary encryption keys for each session, so past data can't be decrypted later by new users
firewall	a security tool (hardware or software) that monitors and filters network traffic to decide what data gets in or out — like a digital bouncer protecting the network

6.5.1 Securing Small Wireless Networks

Guiding Question: How can I tell if a wireless router's settings are secure – and what can I do to fix it if they're not?

Lesson Launch (5 minutes)

Scenario: You just started your job at a small accounting office. Your boss hands you a shiny new Wi-Fi router and says: Can you get this set up before tax season? We need it locked down – clients will be using our Wi-Fi in the lobby, and we don't want anything leaking.

Here's the list of what you're supposed to do:

- Rename the SSID to something neutral.
- Change the default admin password.
- Enable WPA3 encryption.
- Disable WPS.
- Set up a guest network for waiting clients.
- Update the firmware to the latest version.

Say: "Take 30 seconds:

- Which of these steps do you already know how to do – or at least think you understand?
- Which ones sound like tech alphabet soup?"

  Accept all reasonable responses from students.

Say: "You might recognize most of these terms separately, but together, they form a task list that can feel overwhelming. Let's break it down, step by step, and learn how each one helps you secure a small wireless network."

Router Real Talk (30-35 minutes)

- Distribute the Student Handout 6.5.1 – Securing Small Wireless Networks and have students fill in the answers during the lesson.

Getting In: Front Door Settings

- Students should understand that a recognizable ***service set identifier (SSID)*** makes your network easier to target. This setting doesn't protect your data directly, but it can keep your network off the radar.
- Changing the admin password protects your router's setup page. If someone gets here, they can change everything.
- The ***pre-shared key (PSK)*** gets shared with everyone who connects, so it needs to be strong and changed if there's ever a risk.

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Securing the Signal: Encryption and Protection

- We use encryption so other people can't see what we send over Wi-Fi. ***Wi-Fi Protected Access (WPA)*** is the name of the system that manages that encryption. It's been upgraded over time with each version offering better encryption and better features.
- If your devices and router support WPA3, that's what you should use. It has the best protection, including something special called forward secrecy. But if WPA3 causes connection issues, WPA2 is still a good choice.
- ***Forward secrecy*** is one of the coolest things about WPA3. It means that even if someone gets your password later, they can't go back and read things you did before. It's like every session has a key that disappears when you're done.

Match the SSID to the Password

- This matching activity introduces students to examples of well-constructed SSIDs and passphrases that follow best practices: SSIDs are vague but functional, and passwords are made of random words and numbers.
- In this activity, SSIDs and passwords appear thematically matched for the sake of practice—but in reality, they don't have to "match" to be effective.
- One SSID/password pair is intentionally **insecure**, and students will be asked to identify it and explain why.
- The slide is animated to walk through each match from top to bottom, with arrows connecting the SSIDs to their corresponding passwords.
- After the matches are revealed, the animation will highlight the insecure pair – **SmithFamily_WiFi** and **BobAliceJake1999** – by removing arrows and drawing a box around that row.
- This pair is insecure because the SSID includes personally identifiable information (PII) and the password uses what appears to be family member names and a significant year, making it easier for someone to guess.
- While it may be convenient and memorable for the household, this combination could be easily exploited by a malicious actor targeting the network.

SSID & Passphrase Creation Challenge

- Give each student a small slip of scratch paper or an index card.
- Students will create an SSID and password combination that is vague but functional, inspired by the matching game examples. The combination should *describe them without revealing obvious personal details.*
- Let students know not to show their combos to others – part of the fun will be guessing who wrote what.
- Once everyone has submitted their slip, shuffle them and choose 3-5 to read aloud.

- After each one, let the class try to guess whose it might be. If time allows, read more or save extras for early-finisher time.
- If a combination is a little too revealing (like the Smith family example), pause for a **quick, gentle discussion** about what personal info might've been exposed – and how it could be improved.

Locking It Down: Advanced Settings

- ***Wi-Fi Protected Setup (WPS)*** was meant to make connecting easy, but that convenience comes with risk – especially the PIN method. Attackers can guess it. You're better off turning it off and using a strong passphrase instead.
- Think of the ***guest network*** like the guest room of a house. It is the part of the house that you are willing to allow your guest to use because sleeping in your bed, wearing your clothes, and emptying your fridge are not options you want your guest exploring. A separate network for them to stream their videos and check their social sites keeps them from sniffing through your files and using all your printer ink!
- ***Firmware*** isn't something you interact with daily, but it's what makes your router run. Just like you update your phone, your router needs security patches too.
- A ***firewall*** acts like a digital bouncer. It checks traffic coming into your network and stops anything that looks shady. It's not perfect, but it's an easy layer of defense to leave on.

Lab: Wi-Fortify: Secure the Router (15-20 minutes)

Part 1: Securing Small Wireless Match

- Purpose: Students analyze router settings and decide whether they represent secure or insecure configurations. Then, students will examine simple network scenarios to sort which best practice solves the scenario problem.
- How to Run it:
  - This portion is student-paced and self-checking. Direct students to use the sortsecure.html drag-and-drop activities first to check their understanding.
  - After completing the sort and the scenarios, students transfer their answers to the handout's versions.
  - Watch for confusion around settings like WPS or firmware. Those often spark "wait, that's a thing?" moments worth talking about.
- Students should complete this part in about five to seven minutes.

Part 2: Router Simulation

- Purpose: Students walk through a simplified web-based router configuration interface and apply key security settings.
- How to Run it:
  - The simulation mimics a real admin interface but strips away brand-specific clutter.

    - Students use the handout to record settings before and after changes (e.g., firmware version, SSID, WPS status).
    - Encourage students to "think like a technician." They're configuring for a business, not just their own house.
- Let students explore and help each other – but emphasize accuracy. Skipping the firmware update or leaving WPS on could lead to problems in the challenge question later.

Putting It All Together

- Which router setting do you think is the most important for keeping your Wi-Fi secure? Why?
  Student answers may vary, but many may identify the Wi-Fi password or passphrase as the most important. Students may reason that it's the first line of defense against unauthorized access and something within their immediate control.
- Was there a setting that surprised you – either because you didn't expect it to be insecure or didn't know it mattered?
  Student responses may include surprise over WPS (Wi-Fi Protected Setup). Some may not have realized that routers include a button or PIN system that allows access without typing a password — and that this feature can be easily exploited if left enabled.
- If your friend asked you for help making their Wi-Fi more secure, what's one thing from today's lab you'd be confident helping them do?
  Student answers will vary but may include things like changing the SSID or disabling WPS, which may feel approachable and low-risk to adjust. Others might feel confident checking for firmware updates or switching the encryption setting to WPA3.

Putting It All Together (10-15 minutes)

- What setting from today's lesson do you think is most important for keeping a small network secure – and why?
  Student answers may vary but include that changing the default admin password might be the most important because if someone gets into the router settings, they can undo all the other protections. A strong passphrase helps block unauthorized changes.
- A coworker sets their SSID to "LobbyWiFi" and the password to "Accountant2023." What's wrong with this setup, and how could they fix it?
  "LobbyWiFi" gives away the physical location, which could attract attacker nearby. "Accountant2023" is too predictable – it shares job info and a likely year of creation. They should choose a vague SSID and use a stronger password made of unrelated words or use a password manager.
- You want to set up a guest network at your house. What would you name it, what kind of password would you use, and why?
  Student answers may vary. A sample answer could be, "I might call it 'SunroomNet' – something vague but helpful – and use a password like 'CardTacoMoon27.' It's a good password because it uses unrelated words and numbers. I'd also make sure it's different than my main network password, so guests stay separate from my devices."

- Imagine you're helping a friend set up their new router. They say, "I don't care about all that security stuff. I just want it to work." What would you say to convince them that securing their wireless network is worth the effort?

  Student answers may vary. A sample answer could be, "I'd explain that leaving a router unsecured is like leaving your front door unlocked. Anyone nearby could use your internet, snoop on your data, or even change your settings. A few small steps — like using a strong password, turning off WPS, and updating firmware — can stop attacks before they start. It's not just about being "paranoid" — it's about protecting your devices, your data, and your privacy."

- Good fences make good neighbors – and good network segmentation. When you take the time to secure your router, you're protecting your devices, your data, and your peace of mind. Whether it's a guest password or a firmware update, small steps now keep out big problems later.